AP Physics 1 — Unit 7.1

Defining Simple Harmonic Motion (SHM)

Student Notes and In-Class Activities

Big Picture (2–3 minutes)

Purpose of today's lesson:

We are learning _____ and

_____.

By the end of today, you should be able to:

- Describe simple harmonic motion (SHM) in words.
- Identify equilibrium position and restoring force.
- Explain why springs and pendulums can show SHM.
- Sketch and interpret basic graphs related to SHM.

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1. Periodic Motion vs. Simple Harmonic Motion (8 minutes)

Definition:
Periodic motion is motion that:

Examples of periodic motion:

- _____

- _____
- _____

BUT: only the last one is an example of Simple Harmonic Motion (SHM): **motion where an object moves back and forth around an _____. The _____ acting on the object is _____ to its _____ and acts in the _____ direction**

Think–Pair–Share Question:

Does *all* periodic motion count as simple harmonic motion?

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2. Equilibrium Position and Restoring Force (10 minutes)

Equilibrium Position:

Restoring Force:

Key idea:
A restoring force always points:

Example: Block on a Horizontal Spring

Instructions:
Draw the spring–block system and label the equilibrium position.

Questions: Consider a spring attached to a block (ignore gravity and friction).

- If the block is displaced to the right, which direction is the spring force?

- If the block is displaced to the left, which direction is the spring force?

Concept Check

A block is displaced 5 cm to the right of equilibrium.

- Direction of restoring force:

  _____

- Is the net force zero?

  _____

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3. Hooke's Law → Newton's Second Law (15 minutes)

Hooke's Law (write from board):

What each symbol means:

- k: _____
- x: _____

Newton's Second Law (write from board):

Combined equation for SHM (write from board):

Key idea:
Acceleration is proportional to:

Thought Questions

- Where is the acceleration largest?

  _____

- Where is the acceleration zero?

  _____

Sketch: Plot acceleration vs. displacement as you stretch a spring (or after you release the spring)

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4. SHM as a Force Law, Not a Path (5 minutes)

Important clarification:

SHM does not mean 'wiggly motion.' It means a _____ that produces _____.

Complete the table:

System	SHM?	Why / Why not
Spring–mass		
Small-angle pendulum		
Ferris wheel		
Bouncing ball		

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5. Momentum Intuition (7 minutes)

Recall from earlier units:

Questions:

- Where is momentum largest during SHM?

  _____

- Is momentum conserved for the block alone?

  _____

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6. Pendulum as SHM (Small Angles Only) (10 minutes)

Draw a free-body diagram of a pendulum bob:

Write the restoring torque relationship (from board):

Connection to springs:

- Spring displacement corresponds to:

  _____

- Spring force corresponds to:

  _____

Concept Check:

If the angular displacement is doubled (small angle), the restoring torque:

_____

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7. Qualitative Graphing (8 minutes)

Force vs. Displacement

Acceleration vs. Displacement

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8. Synthesis & Looking Ahead (5 minutes)

Key ideas from today:

Looking ahead:
Next class we will:

Exit Question:

If the displacement from equilibrium is doubled, what happens to the restoring force?